

Unit 3 — Transport Layer MCQs

Filtered Question Bank — Duplicates Removed

ORIGINAL QUESTIONS

316

DUPLICATES REMOVED

187

UNIQUE QUESTIONS

129

1. What happens in the TCP connection termination phase after the FIN packet is sent?

- A) The data transmission continues until all data is received.
- B) The sender waits for a timeout.
- C) The connection is immediately closed.
- D) An ACK is sent back to acknowledge the FIN.

Answer D) An ACK is sent back to acknowledge the FIN.

2. In TCP, what does the Slow Start algorithm initially do to the congestion window size?

- A) Decreases it exponentially
- B) Keeps it constant
- C) Increases it linearly
- D) Increases it exponentially

Answer D) Increases it exponentially

3. Which of the following is a connectionless protocol at the Transport Layer?

- A) UDP
- B) HTTP
- C) TCP
- D) IP

Answer A) UDP

4. What action does TCP take if it receives an acknowledgment for a packet but later detects that the packet was lost?

- A) Close the connection
- B) Reset the connection
- C) Retransmit the lost packet
- D) Ignore the acknowledgment

Answer C) Retransmit the lost packet

5. What is the primary advantage of the Selective Repeat protocol over Go-Back-N?

- A) It is faster than Go-Back-N.
- B) It sends all packets without waiting for acknowledgments.
- C) It does not require acknowledgments.
- D) It retransmits only the erroneous or lost packets.

Answer D) It retransmits only the erroneous or lost packets.

6. What is the primary function of the Transport Layer in the OSI model?

- A) Ensuring end-to-end communication between devices(process)
- B) Data encryption and decryption
- C) Data Link Layer functionality
- D) Routing of packets

Answer A) Ensuring end-to-end communication between devices(process)

7. Which of the following is true about Selective Repeat?

- A) It does not support pipelining.
- B) It requires the receiver to acknowledge every packet.
- C) It retransmits all packets after a loss.
- D) It only retransmits the lost packets.

Answer D) It only retransmits the lost packets.

8. Which feature of Selective Repeat distinguishes it from Go-Back-N in terms of retransmission?

- A) Selective Repeat does not require acknowledgments.
- B) Selective Repeat uses cumulative acknowledgments.
- C) Selective Repeat retransmits only individual lost packets.
- D) Selective Repeat retransmits all packets in the window.

Answer C) Selective Repeat retransmits only individual lost packets.

9. In a pipelined protocol, what does the term "window size" refer to?

- A) The time interval for packet transmission
- B) The size of the buffer at the receiver
- C) The size of the data segment
- D) The number of packets that can be sent without waiting for an acknowledgment

Answer D) The number of packets that can be sent without waiting for an acknowledgment

10. Which rdt version improves upon previous versions by using sequence numbers to handle packet loss and duplication?

- A) rdt 2.1
- B) rdt 3.0
- C) rdt 1.0
- D) rdt 2.0

Answer A) rdt 2.1

11. In Go-Back-N protocol, what happens when a packet is lost?

- A) The connection is reset
- B) The data is discarded
- C) Only the lost packet is retransmitted
- D) All packets from the lost packet onward are retransmitted

Answer D) All packets from the lost packet onward are retransmitted

12. Which pipelining protocol requires the receiver to send acknowledgments only for out-of-order packets?

- A) Selective Repeat
- B) rdt 3.0
- C) Stop-and-Wait
- D) Go-Back-N

Answer A) Selective Repeat

13. Which of the following is a limitation of UDP compared to TCP?

- A) Lack of encryption
- B) Lack of connection establishment
- C) Lack of flow control
- D) Lack of checksum

Answer B) Lack of connection establishment

14. Which of the following best describes the Checksum in UDP?

- A) It establishes a connection between two devices.
- B) It manages flow control between devices.
- C) It is used for error correction.
- D) It is used to ensure data integrity.

Answer D) It is used to ensure data integrity.

15. In UDP, what action does the sender take upon receiving no acknowledgment for a transmitted packet?

- A) Sends a request for acknowledgment
- B) Does nothing (no retransmission)
- C) Retransmits the packet
- D) Waits for a longer time

Answer B) Does nothing (no retransmission)

16. In the context of multiplexing, what does the term "demultiplexing" refer to?

- A) Directing data to the correct application process based on port numbers
- B) Encrypting data for secure transmission
- C) Combining multiple signals into one
- D) Route data to the proper destination network

Answer A) Directing data to the correct application process based on port numbers

17. What is a key benefit of Selective Repeat over Go-Back-N in terms of efficiency?

- A) Lower memory usage
- B) Faster data transfer rates
- C) Reduced retransmission of non-lost packets
- D) Simpler error handling

Answer C) Reduced retransmission of non-lost packets

18. In UDP, what is the consequence of the receiver receiving a corrupted packet?

- A) The receiver sends a NACK to the sender.
- B) The packet is corrected and processed.
- C) The receiver requests a retransmission.
- D) The packet is silently discarded.

Answer D) The packet is silently discarded.

19. Which of the following is true about the Go-Back-N protocol?

- A) It uses selective acknowledgments for each packet.
- B) It requires the receiver to store out-of-order packets.
- C) It can only send one packet at a time.
- D) It can send multiple packets without waiting for an acknowledgment.

Answer D) It can send multiple packets without waiting for an acknowledgment.

20. What new feature does rdt 2.1 introduce compared to rdt 2.0?

- A) Improved error handling
- B) Sequence numbers
- C) Negative acknowledgments
- D) Timeout mechanisms

Answer A) Improved error handling

21. Which TCP feature helps in detecting lost packets and ensuring reliable data transfer?

- A) Flow control
- B) Congestion control
- C) Sequence numbers
- D) Checksum

Answer C) Sequence numbers

22. What type of acknowledgment in pipelined protocol does TCP use to confirm the receipt of data?

- A) Cumulative Acknowledgment
- B) Duplicate Acknowledgment
- C) Selective Acknowledgment (SACK)
- D) Negative Acknowledgment (NACK)

Answer A) Cumulative Acknowledgment

23. In Selective Repeat, how is the problem of packet loss addressed compared to Go-Back-N?

- A) The entire window of packets is retransmitted
- B) Packets are discarded
- C) Only the lost packet is retransmitted
- D) All packets from the lost packet onward are retransmitted

Answer C) Only the lost packet is retransmitted

24. Which protocol feature in TCP allows the sender to infer the occurrence of packet loss without waiting for a timeout?

- A) Fast Retransmit
- B) Delayed Acknowledgment
- C) Selective Acknowledgment (SACK)
- D) Duplicate Acknowledgments

Answer A) Fast Retransmit

25. How does TCP react to detecting network congestion?

- A) It establishes a new connection
- B) It skips data acknowledgment
- C) It decreases the congestion window size
- D) It increases the transmission rate

Answer C) It decreases the congestion window size

26. Which reliable data transfer protocol includes positive and negative acknowledgments?

- A) rdt 2.2
- B) rdt 3.0
- C) rdt 1.0
- D) rdt 2.0

Answer D) rdt 2.0

27. Which of the following UDP characteristics makes it unsuitable for applications that require guaranteed delivery of data?

- A) No error correction capability
- B) All Options
- C) Lack of connection establishment
- D) No flow control mechanism

Answer B) All Options

28. What happens to the TCP congestion window during the slow start phase?

- A) It decreases linearly
- B) It is reset
- C) It increases exponentially with each acknowledgment
- D) It remains constant until a timeout occurs

Answer C) It increases exponentially with each acknowledgment

29. How does TCP's Fast Retransmit mechanism improve performance during packet loss?

- A) By increasing the congestion window exponentially
- B) By sending an acknowledgment for every received packet
- C) By immediately retransmitting a packet upon receiving three duplicate acknowledgments
- D) By waiting for a timeout before retransmitting a packet

Answer C) By immediately retransmitting a packet upon receiving three duplicate acknowledgments

30. Which component of the Transport Layer is responsible for reassembling segments into the correct order at the destination?

- A) Sequence Number
- B) Checksum
- C) Multiplexer
- D) Demultiplexer

Answer A) Sequence Number

31. Which of the following best describes how multiplexing works in connection-oriented protocols?

- A) It separates data streams into individual connections
- B) It combines multiple streams of data into a single connection before transmission
- C) It compresses data to reduce bandwidth usage
- D) It encrypts data for transmission

Answer B) It combines multiple streams of data into a single connection before transmission

32. What is the key characteristic of rdt 1.0?

- A) It is a reliable data transfer protocol without any error detection.
- B) It handles packet loss.
- C) It handles bit errors using acknowledgments.
- D) It includes a checksum for error detection.

Answer A) It is a reliable data transfer protocol without any error detection.

33. Which scenario in TCP triggers the transition from Congestion Avoidance to Slow Start?

- A) Upon the receipt of a duplicate acknowledgment
- B) When the congestion window size exceeds the threshold
- C) Upon the receipt of the first acknowledgment
- D) When the timeout occurs for an unacknowledged packet

Answer D) When the timeout occurs for an unacknowledged packet

34. In TCP, what is the purpose of the three-way handshake?

- A) To negotiate encryption methods
- B) To establish a reliable connection
- C) To avoid network congestion
- D) To manage flow control

Answer B) To establish a reliable connection

35. What new feature is introduced in rdt 3.0 compared to rdt 2.0?

- A) Error detection
- B) Timeout-based retransmission
- C) Negative acknowledgment
- D) Acknowledgment-based retransmission

Answer B) Timeout-based retransmission

36. Which pipelined protocol is better suited for environments with high packet loss?

- A) Selective Repeat
- B) Go-Back-N
- C) Neither is suited
- D) Both are equally suited

Answer A) Selective Repeat

37. Which rdt version first introduces the concept of sequence numbers to detect duplicate packets?

- A) rdt 2.0
- B) rdt 1.0
- C) rdt 3.0
- D) rdt 2.1

Answer D) rdt 2.1

38. What role does the checksum play in UDP?

- A) To verify data integrity
- B) To encrypt data
- C) To establish connections
- D) To manage flow control

Answer A) To verify data integrity

39. What triggers the end of the slow start phase in TCP?

- A) The congestion window size exceeds a predefined threshold
- B) The data rate is adjusted
- C) The connection is closed
- D) A timeout occurs

Answer A) The congestion window size exceeds a predefined threshold

40. What is the key challenge addressed by the rdt 3.0 protocol?

- A) Bit errors
- B) Flow control
- C) Packet loss
- D) Duplicate packets

Answer C) Packet loss

41. In rdt 3.0, what mechanism is introduced to handle packet loss in addition to error detection?

- A) Sequence Numbers
- B) Flow Control
- C) Timeouts and Retransmission
- D) Checksum

Answer C) Timeouts and Retransmission

42. Which of the following correctly describes the rdt 2.1 protocol?

- A) It is a connectionless protocol.
- B) It can handle only packet loss.
- C) It introduces the concept of pipelining.
- D) It can handle bit errors and uses checksums.

Answer D) It can handle bit errors and uses checksums.

43. What does UDP stand for?

- A) User Data Protection
- B) Universal Data Protocol
- C) Unidirectional Data Protocol
- D) User Datagram Protocol

Answer D) User Datagram Protocol

44. What is the key feature of the Stop-and-Wait protocol?

- A) It is a connectionless protocol.
- B) It allows the sender to send multiple packets before receiving an acknowledgment.
- C) It uses a sliding window for flow control.
- D) It sends one packet and waits for an acknowledgment before sending the next one.

Answer D) It sends one packet and waits for an acknowledgment before sending the next one.

45. Which TCP field indicates the position of the first byte of data in the segment?

- A) Urgent Pointer
- B) Acknowledgment Number
- C) Sequence Number
- D) Window Size

Answer C) Sequence Number

46. What is the purpose of flow control in the Transport Layer?

- A) To control the speed of data transmission between sender and receiver
- B) To translate IP addresses to MAC addresses
- C) To determine the best route for data packets
- D) To encrypt data before transmission

Answer A) To control the speed of data transmission between sender and receiver

47. Which TCP feature allows it to handle out-of-order packets?

- A) Sequence numbers
- B) Checksum
- C) Congestion control
- D) Flow control

Answer A) Sequence numbers

48. Which mechanism helps TCP avoid network congestion?

- A) Error Detection
- B) Slow Start
- C) Fast Retransmit
- D) Congestion Avoidance

Answer D) Congestion Avoidance

49. What is the primary role of the TCP three-way handshake?

- A) To compress data before transmission
- B) To encrypt data
- C) To synchronize sequence numbers between sender and receiver
- D) To establish a secure connection

Answer C) To synchronize sequence numbers between sender and receiver

50. Which field in a UDP packet is used to detect errors in the transmitted data?

- A) Data
- B) Checksum
- C) Length
- D) Port number

Answer B) Checksum

51. What does the Selective Repeat protocol do with out-of-order packets?

- A) It immediately forwards them to the application layer.
- B) It requests retransmission of all packets.
- C) It stores them until the missing packets are received.
- D) It discards them.

Answer C) It stores them until the missing packets are received.

52. What is the default port number for HTTP, a protocol that uses TCP?

- A) 443
- B) 80
- C) 21
- D) 20

Answer B) 80

53. What is the role of the ACK flag in TCP?

- A) To indicate the beginning of data transmission
- B) To acknowledge the receipt of data
- C) To terminate a connection
- D) To initiate a connection

Answer B) To acknowledge the receipt of data

54. What happens if the UDP checksum does not match the data received?

- A) The connection is reset
- B) The packet is processed with a warning
- C) The packet is retransmitted
- D) The packet is discarded

Answer D) The packet is discarded

55. Which of the following protocols does not provide flow control?

- A) HTTP
- B) SCTP
- C) UDP
- D) TCP

Answer C) UDP

56. What mechanism does TCP use to adjust the congestion window in response to network conditions?

- A) Error Detection
- B) Congestion Avoidance
- C) Fast Retransmit
- D) Slow Start

Answer B) Congestion Avoidance

57. In the context of UDP, what happens if a packet with a corrupted checksum is received?

- A) The packet is corrected and processed.
- B) The receiver requests retransmission.
- C) The receiver sends a negative acknowledgment.
- D) It is discarded silently.

Answer D) It is discarded silently.

58. How does TCP ensure that data is received correctly and in order?

- A) By establishing multiple connections
- B) By using error detection codes
- C) By using encryption
- D) By using sequence numbers and acknowledgments

Answer D) By using sequence numbers and acknowledgments

59. In a connectionless protocol, how is data routed to the correct application?

- A) By connection handshaking
- B) By data encryption
- C) By using port numbers
- D) Through connection establishment

Answer C) By using port numbers

60. What role does the congestion window (cwnd) play in TCP's Congestion Control mechanism?

- A) It limits the number of segments that can be sent without acknowledgment.
- B) It ensures the correct order of data delivery.
- C) It manages the retransmission of lost packets.
- D) It specifies the maximum segment size.

Answer A) It limits the number of segments that can be sent without acknowledgment.

61. In which rdt protocol is the Stop-and-Wait mechanism used for flow control?

- A) rdt 1.0
- B) rdt 3.0
- C) rdt 2.1
- D) rdt 2.0

Answer B) rdt 3.0

62. In the context of the Transport Layer, what does multiplexing refer to?

- A) Splitting data into multiple streams
- B) Encrypting data before transmission
- C) Routing data to different destinations
- D) Combining multiple signals into one signal

Answer D) Combining multiple signals into one signal

63. What is the role of sequence numbers in TCP?

- A) To keep track of the order of packets
- B) To detect errors
- C) To manage flow control
- D) To establish a connection

Answer A) To keep track of the order of packets

64. What is the primary function of port numbers in connectionless multiplexing?

- A) To identify the destination application for each data packet
- B) To ensure secure data transfer
- C) To compress data for faster transmission
- D) To establish a connection between sender and receiver

Answer A) To identify the destination application for each data packet

65. What is the primary function of the UDP length field in its header?

- A) To specify the size of the data segment
- B) To specify the window size
- C) To detect errors in the data
- D) To indicate the number of segments

Answer A) To specify the size of the data segment

66. Which protocol does not require acknowledgment for the received data?

- A) Selective Repeat
- B) TCP
- C) rdt 2.0
- D) UDP

Answer D) UDP

67. Which field in the TCP header is responsible for ensuring data arrives in the correct sequence?

- A) Checksum
- B) Urgent Pointer
- C) Window Size
- D) Sequence Number

Answer D) Sequence Number

68. Which of the following is true for rdt 3.0 but not for rdt 2.2?

- A) It relies on sequence numbers for duplicate detection
- B) It uses acknowledgments for error recovery
- C) It introduces timers to manage retransmissions.
- D) It can handle packet loss as well as bit errors.

Answer C) It introduces timers to manage retransmissions.

69. Which protocol improves upon Go-Back-N by reducing the number of retransmissions?

- A) rdt 3.0
- B) Stop-and-Wait
- C) rdt 2.1
- D) Selective Repeat

Answer D) Selective Repeat

70. Which of the following best describes the TCP congestion control mechanism?

- A) It adjusts the congestion window based on network conditions.
- B) It uses a fixed window size for flow control.
- C) It only operates at the Application Layer.
- D) It ignores network congestion and sends data continuously.

Answer A) It adjusts the congestion window based on network conditions

71. Which rdt version includes a timeout mechanism for retransmissions?

- A) rdt 2.0
- B) rdt 1.0
- C) rdt 3.0
- D) rdt 2.1

Answer C) rdt 3.0

72. What happens if the UDP checksum does not match the data received?

- A) The packet is retransmitted.
- B) The packet is discarded.
- C) The connection is reset.
- D) The packet is processed with a warning.

Answer B) The packet is discarded

73. What is the main purpose of the sequence number in TCP segments?

- A) To ensure data is delivered in order
- B) To identify the sender and receiver
- C) To calculate the checksum
- D) To detect duplicate packets

Answer A) To ensure data is delivered in order

74. Which transport layer protocol is preferred for applications that require fast transmission but can tolerate some data loss?

- A) UDP
- B) TCP
- C) ARP
- D) ICMP

Answer A) UDP

75. What characteristic of UDP makes it suitable for applications like online gaming and live streaming?

- A) Error recovery
- B) Reliable delivery
- C) Data ordering
- D) Low latency

Answer D) Low latency

76. Which protocol at the Transport Layer does not guarantee delivery of data?

- A) HTTP
- B) TCP
- C) FTP
- D) UDP

Answer D) UDP

77. What is the primary purpose of rdt 3.0 in reliable data transfer protocols?

- A) To manage flow control between sender and receiver
- B) To handle bit errors and packet loss
- C) To encrypt data for secure transmission
- D) To establish a connection between two devices

Answer B) To handle bit errors and packet loss

78. What is the initial value of the congestion window (cwnd) in TCP Slow Start?

- A) 1 segment
- B) 1 byte
- C) 4 segments
- D) 2 segments

Answer A) 1 segment

79. In a multiplexing context, how does the Transport Layer differentiate between various data streams?

- A) By using sequence numbers
- B) By using IP addresses
- C) By using MAC addresses numbers
- D) By using port numbers

Answer D) By using port numbers

80. What does the Transport Layer use to differentiate multiple services running on the same host?

- A) MAC Addresses
- B) IP Addresses
- C) Sequence Numbers
- D) Port Numbers

Answer D) Port Numbers

81. Which of the following describes the Go-Back-N protocol's retransmission strategy?

- A) No retransmission is performed
- B) Retransmit only the last packet
- C) Retransmit all unacknowledged packets starting from the lost packet
- D) Retransmit only the lost packet

Answer C) Retransmit all unacknowledged packets starting from the lost packet

82. Which aspect of UDP makes it less reliable than TCP?

- A) Ordered data delivery
- B) Error detection
- C) Use of checksums
- D) Lack of connection establishment

Answer D) Lack of connection establishment (Note: Lack of guaranteed ordered delivery is a factor, but among options, "Lack of connection establishment" / no reliability guarantees is standard).

83. Which field in the UDP header is used to verify data integrity?

- A) Checksum
- B) Length
- C) Destination Port
- D) Source Port

Answer A) Checksum

84. Which of the following is a limitation of UDP compared to TCP?

- A) Lack of flow control
- B) Lack of checksum
- C) connection establishment
- D) Lack of encryption

Answer A) Lack of flow control

85. In TCP, what happens if a segment is received with an incorrect checksum?

- A) The segment is discarded
- B) The segment is accepted with a warning
- C) The connection is closed
- D) The segment is retransmitted

Answer A) The segment is discarded

86. Which rdt protocol introduces the concept of acknowledgments with sequence numbers to detect duplicate packets?

- A) rdt 2.0
- B) rdt 1.0
- C) rdt 3.0
- D) rdt 2.1

Answer D) rdt 2.1

87. How does TCP ensure that no data is lost during transmission in case of packet loss?

- A) By using sequence numbers and acknowledgments
- B) By using a checksum for error detection
- C) By sending duplicate packets
- D) By relying on lower layers for error correction

Answer A) By using sequence numbers and acknowledgments

88. What happens to incoming data in connection-oriented multiplexing?

- A) It is routed to the correct application based on port numbers
- B) It is combined into a single data stream
- C) It is discarded if not acknowledged
- D) It is encrypted

Answer A) It is routed to the correct application based on port numbers

89. Which Transport Layer protocol is most suitable for real-time applications like video streaming?

- A) FTP
- B) TCP
- C) SMTP
- D) UDP

Answer D) UDP

90. Which mechanism in TCP is responsible for avoiding network congestion?

- A) Checksum
- B) Flow Control
- C) Slow Start
- D) Multiplexing

Answer C) Slow Start

91. Which pipelined protocol is more efficient in handling retransmissions in case of packet loss?

- A) Neither is efficient
- B) Selective Repeat
- C) Go-Back-N
- D) Both are equally efficient

Answer B) Selective Repeat

92. Which feature of Selective Repeat distinguishes it from Go-Back-N in terms of retransmission?

- A) Selective Repeat retransmits only the lost or corrupted packets.
- B) Selective Repeat retransmits all packets after a loss
- C) Selective Repeat uses cumulative acknowledgments.
- D) Selective Repeat does not require acknowledgments.

Answer A) Selective Repeat retransmits only the lost or corrupted packets.

93. What is the main advantage of connection-oriented multiplexing?

- A) It increases the speed of data transfer
- B) It reduces the amount of data transmitted
- C) It ensures that data is correctly delivered to the appropriate application
- D) It simplifies the data transfer process

Answer C) It ensures that data is correctly delivered to the appropriate application

94. What is the primary function of the Transport Layer in the OSI model?

- A) Data encryption and decryption
- B) Routing of packets
- C) Data Link Layer functionality
- D) Ensuring end-to-end communication between devices (process)

Answer D) Ensuring end-to-end communication between devices(process)

95. Which reliable data transfer protocol includes mechanisms for handling bit errors?

- A) rdt 1.0
- B) rdt 2.0
- C) rdt 3.0
- D) None of the above

Answer B) rdt 2.0

96. Which of the following best describes the TCP congestion control mechanism?

- A) It uses a fixed window size for flow control.
- B) It adjusts the congestion window based on network conditions.
- C) It only operates at the Application Layer.
- D) It ignores network congestion and sends data continuously

Answer B) It adjusts the congestion window based on network conditions.

97. How does rdt 3.0 handle lost packets more effectively than previous versions?

- A) By using timeouts and retransmissions
- B) using positive and negative acknowledgments
- C) By introducing multiple acknowledgment messages
- D) By implementing sequence numbers

Answer A) By using timeouts and retransmissions

98. In the context of multiplexing, what does the term "demultiplexing" refer to?

- A) Combining multiple signals into one
- B) Splitting one signal into multiple channels
- C) Encrypting data for secure transmission
- D) Directing data to the correct application process based on port numbers

Answer D) Directing data to the correct application process based on port numbers

99. What is the result of TCP's congestion window size reaching the threshold during the slow start phase?

- A) Transition to congestion avoidance phase
- B) Immediate retransmission of lost packets
- C) Doubling of the window size
- D) Termination of the connection

Answer A) Transition to congestion avoidance phase

100. Which mechanism in TCP ensures that the sender does not overwhelm the receiver's buffer capacity?

- A) Congestion Control
- B) Flow Control
- C) Sequence Numbers
- D) Three-Way Handshake

Answer B) Flow Control

101. What is the default subnet mask for a Class C IP address?

- A) 255.255.255.255
- B) 255.255.0.0
- C) 255.255.255.0
- D) 255.0.0.0

Answer C) 255.255.255.0

102. Which of the following is an advantage of IPv6 over IPv4?

- A) All Options
- B) Larger address space
- C) Simplified header structure
- D) No need for NAT

Answer A) All Options

103. What is the subnet address of subnet C (CIDR notation)?

- A) 172.27.182.0/25
- B) 172.27.182.0/26
- C) 172.27.182.0/24
- D) 172.27.182.0/27

Answer A) 172.27.182.0/25

104. Which of the following is a key function of the Network Layer?

- A) Data Encryption
- B) Routing
- C) Session Management
- D) Flow Control

Answer B) Routing

105. What is the maximum number of hosts available in a subnet with a /22 prefix?

- A) 4094
- B) 1022
- C) 2046
- D) 510

Answer B) 1022

106. Compute the Internet checksum value for the two 16-bit binary words (1100001000111110 and 11000000 00000010):

- A) 01111001 10111110
- B) 11000000 00000010
- C) 10000110 01000001
- D) 11000110 00111110

Answer C) 10000110 01000001

107. In classful IP addressing, what is the range of Class B addresses?

- A) 224.0.0.0 to 239.255.255.255
- B) 192.0.0.0 to 223.255.255.255
- C) 0.0.0.0 to 127.255.255.255
- D) 128.0.0.0 to 191.255.255.255

Answer D) 128.0.0.0 to 191.255.255.255

108. In the context of TCP, what is the primary function of flow control?

- A) To manage congestion in the network
- B) To ensure data integrity
- C) To encrypt data before transmission
- D) To prevent the sender from overwhelming the receiver

Answer D) To prevent the sender from overwhelming the receiver

109. In UDP, what is the maximum possible value of the checksum field?

- A) 0xFF00
- B) 0x00FF
- C) 0xFFFF
- D) 0xFFFFF

Answer C) 0xFFFF

110. In TCP, which algorithm is used to probe the network for the available bandwidth by increasing the congestion window size exponentially?

- A) Fast Retransmit
- B) Congestion Avoidance
- C) Fast Recovery
- D) Slow Start

Answer D) Slow Start

111. Which reliable data transfer protocol includes mechanisms for handling bit errors?

- A) rdt 2.1
- B) rdt 2.0
- C) rdt 3.0
- D) None of the above

Answer B) rdt 2.0

112. Which feature of Selective Repeat distinguishes it from Go-Back-N in terms of retransmission?

- A) Selective Repeat uses cumulative acknowledgments.
- B) Selective Repeat retransmits only the lost or corrupted packets.
- C) Selective Repeat retransmits all packets after a loss.
- D) Selective Repeat does not require acknowledgments.

Answer B) Selective Repeat retransmits only the lost or corrupted packets.

113. Which of the following is a random access protocol?

- A) Slotted Aloha
- B) FDMA
- C) Token Passing
- D) TDMA

Answer A) Slotted Aloha

114. Which of the following is a feature of the Ethernet protocol?

- A) CSMA/CD
- B) Polling
- C) Time Division Multiplexing
- D) Token Passing

Answer A) CSMA/CD

115. What is the primary function of a router in a network?

- A) To establish a connection between devices
- B) To resolve IP addresses to MAC addresses
- C) To detect and correct errors
- D) To route data between networks

Answer D) To route data between networks

116. What is the main purpose of bit stuffing?

- A) Collision detection
- B) Frame synchronization
- C) Data compression
- D) Error correction

Answer B) Frame synchronization

117. Which type of Ethernet uses CSMA/CD as the access method?

- A) 1000Base-T
- B) 100Base-TX
- C) All Options
- D) 10Base-T

Answer C) All Options

118. Which of the following protocols uses a pre-defined time slot for each station to transmit data?

- A) CDMA
- B) TDMA
- C) CSMA/CD
- D) ALOHA

Answer B) TDMA

119. In a pipelined protocol like Go-Back-N, what happens when a packet is lost?

- A) The receiver discards all packets until the lost packet is received.
- B) The receiver sends an acknowledgment for the next packet.
- C) The sender retransmits the lost packet and all subsequent packets
- D) The connection is terminated.

Answer C) The sender retransmits the lost packet and all subsequent packets

120. In TCP, what triggers the transition from Slow Start to Congestion Avoidance?

- A) The expiration of a timer
- B) A lost packet
- C) Reaching the threshold value for the congestion window
- D) The receipt of a duplicate acknowledgment

Answer C) Reaching the threshold value for the congestion window

121. How does TCP ensure that data packets are delivered in the correct sequence?

- A) By using multiple connections
- B) By using sequence numbers and reordering packets
- C) By using ack and reordering packets
- D) By encrypting the data

Answer B) By using sequence numbers and reordering packets

122. What does demultiplexing do in a connection-oriented transport protocol?

- A) Encrypts data
- B) Combines multiple streams of data
- C) Establishes a connection
- D) Separates data into different application channels

Answer D) Separates data into different application channels

123. What is the main difference between rdt 2.1 and rdt 2.2?

- A) rdt 2.2 handles duplicate packets better than rdt 2.1.
- B) rdt 2.2 introduces the use of timers, while rdt 2.1 does not.
- C) rdt 2.1 is a connectionless protocol, while rdt 2.2 is connection-oriented.
- D) rdt 2.1 uses negative acknowledgments, while rdt 2.2 does not.

Answer D) rdt 2.1 uses negative acknowledgments, while rdt 2.2 does not.

124. In a multiplexing context, how does the Transport Layer differentiate between various data streams?

- A) By using IP addresses
- B) By using port numbers
- C) By using MAC addresses
- D) By using sequence numbers

Answer B) By using port numbers

125. What is the key difference between TCP and UDP?

- A) TCP is faster than UDP.
- B) TCP is connection-oriented, while UDP is connectionless.
- C) UDP operates at the Network Layer, while TCP operates at the Transport Layer.
- D) UDP provides reliable data transfer, while TCP does not.

Answer B) TCP is connection-oriented, while UDP is connectionless.

126. Which of the following transport layer protocols is known for providing a reliable end-to-end communication?

- A) ARP
- B) UDP
- C) ICMP
- D) TCP

Answer D) TCP

127. Which reliable data transfer protocol includes mechanisms for handling bit errors?

- A) rdt 2.0
- B) rdt 2.1
- C) rdt 1.0
- D) rdt 3.0

Answer A) rdt 2.0

128. In which scenarios is TCP less suitable compared to UDP?

- A) Applications with high-security requirements
- B) Applications requiring high reliability and order
- C) Applications with large data transfers
- D) Applications where speed is critical and occasional data loss is acceptable

Answer D) Applications where speed is critical and occasional data loss is acceptable

129. Which mechanism does TCP use to control the flow of data between sender and receiver when the receiver's buffer is full?

- A) Sliding Window
- B) Flow Control
- C) Congestion Control
- D) Checksum

Answer B) Flow Control